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Mitigating the transfer risk of antibiotic resistance genes from fertilized soil to cherry radish during the application of insect fertilizer☆

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A B S T R A C T

The transfer of antibiotic resistance genes (ARGs) from fertilized soil to vegetables, particularly those consumed raw, causes significant public health risks through the food chain. Black soldier fly larvae can efficiently convert animal manure into organic fertilizer with reduced antibiotic resistance. This study utilized metagenomic sequencing to investigate fields treated with control organic fertilizer (COF), black soldier fly organic fertilizer (BOF), and no fertilizer, with the aim of assessing the transfer risks of ARGs from soil to cherry radish. The results indicated that BOF significantly reduced the richness and abundance of ARGs in both soil and cherry radish compared to COF, reducing 13 ARG subtypes and a 27.6% decrease in ARG abundance in cherry radish. Moreover, a significant positive correlation was observed between mobile genetic elements (MGEs) and virulence factors (VFs) with ARGs, with BOF treatment resulting in a relative abundance reduction of 32.8% and 29.1%, respectively. The complexity of networks involving ARGs with MGEs, VFs, and microbial communities in the BOF treatment was 54.2%, 32.3%, and 32.8% lower, respectively, than the COF treatment. Further analysis of metagenomic-assembled genomes (MAGs) revealed the co-occurrence of ARGs, MGEs, and VFs in cherry radish, indicating the presence of potential pathogenic antibiotic-resistant bacteria (PARB). Notably, the abundance of these PARB in BOF radishes decreased by 45.6% compared to COF. These findings underscore the efficacy of insect fertilizer in mitigating the transfer risks of ARGs to radish, highlighting the significance of sustainable agricultural practices in managing the environmental and health risks associated with ARGs.

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1. Introduction

The dissemination and accumulation of antibiotic resistance genes (ARGs) in the environment pose a sustainable threat to public health, attracting global attention. Researchers have identified a high diversity and abundance of ARGs in various types of animal manures, including pig manure (Tu et al., 2023, Zhu et al., 2013), chicken manure (Qian et al., 2018), cow manure (Zalewska et al., 2024), and duck manure (Liu et al., 2024). These manures, which are frequently utilized as organic fertilizers in agricultural fields without adequate treatment or following only basic composting, may facilitate the migration of ARGs and antibiotic-resistant bacteria (ARB) into the soil and potentially into the food chain, thereby presenting a significant risk to human health.

Numerous studies have demonstrated that the application of animal manure significantly enhances the richness and abundance of ARGs in soil. For example, Chen et al. (2016) demonstrated that the continuous application of chicken manure organic fertilizer significantly enhanced the levels of β-lactams, tetracyclines, and multi-drug resistance genes in soil, with the enrichment of the *mexF* gene exhibiting an extraordinary increase of 3845-fold. Furthermore, ARGs present in the soil possess the potential to migrate into plant tissues. The abundance of ARGs diminishes from soil to fruit and legume crops, constituting only 0.1 to 0.01 % of their concentration in the soil. Fertilization has been identified as the primary driver of the presence of ARGs in the edible portions of crops (Cerqueira et al., 2019). Notably, ARGs present in agricultural crops can also be transmitted through the food chain to animal populations. For instance, research has demonstrated that the expression of the *blaTEM gene* in the colons of C57BL/6 mice significantly increased following the ingestion of a grape suspension. This finding underscores the potential hazards that ARGs in the food supply pose to animal health (Wang et al., 2024a).

The bio-conversion of animal manure by black soldier fly (BSF) larvae yields bio-organic fertilizer (BOF) with significantly diminished levels of ARGs, MGEs, ARB and transferable ARB (Cai et al., 2018a; Niu et al., 2022; Zhao et al., 2023a,b). Studies have demonstrated that the application of BOF significantly reduces the abundance of ARGs in fertilized soils compared to control conditions (Zhao et al., 2024). Nevertheless, the impact of BOF application on ARGs in vegetables remains inadequately understood. Specifically, cherry radish demonstrates increased susceptibility to the accumulation of ARGs due to extended exposure to soil throughout its growth cycle (Ransirini et al., 2024). Additionally, the prevalence of raw cherry radish consumption raises concerns regarding the potential transfer of ARGs to humans which is not mitigated by the high-temperature cooking processes typically employed.

Therefore, this study utilizes cherry radish as a model organism and employs BSF larvae to bio-convert chicken manure into bio-organic fertilizer (BOF) for field experiments. The primary objective is to examine the impact of BOF application on the abundance of ARGs in both the rhizosphere soil and cherry radish. Additionally, metagenomic sequencing and metagenome-assembled genome (MAG) analysis will be employed to elucidate the co-occurrence of ARGs, mobile genetic elements (MGEs), and virulence factors (VFs) within the radish. The specific objectives of this study are to determine: (1) the diversity and abundance of ARGs in rhizosphere soil and cherry radish following the application of BOF; (2) the transfer risks of ARGs from fertilized soil to cherry radish; (3) the abundance of MGEs and VFs in cherry radish, as well as their co-occurrence with ARGs; (4) the composition of bacterial and fungal communities in cherry radish; and (5) the occurrence risks of pathogenic antibiotic-resistant bacteria (PARB) in cherry radish.

2. Materials and methods

2.1. BSF larvae and chicken manure

The BSF larvae utilized in this study were obtained from a stable BSF population maintained by the Microbiology and Edible Insect Laboratory at Huazhong Agricultural University. Six-day-old BSF larvae were selected for the bio-conversion of chicken manure. Fresh chicken manure was collected from the poultry farm at Huazhong Agricultural University.

2.2. Production of organic fertilizer

The production of BOF was conducted in accordance with our previous studies (Zhao et al., 2024). In brief, a total of 100,000 six-day-old BSF larvae were utilized to bio-convert 100 kg of chicken manure over a period of 12 days. Following the removal of the larvae through sieving, the remaining residues were subjected to composting for an additional 16 days resulting in the formation of BOF. Meanwhile, BSF microbes agents organic fertilizer (BMOF) was made by incorporating 1 % (w/v) of fermented liquid from *Candida rugosa* YE during the BSF conversion stage and 1 % (w/v) of fermented liquid from *Bacillus siamensis* zz07 during the composting stage. These two microbial strains have been demonstrated to possess potential for the reduction of ARGs (Zhao et al., 2024). Additionally, a composting treatment without BSF larvae was established to to produce control organic fertilizer (COF).

2.3. Field experiment design and sampling

The field experiment was conducted in the experimental field of Huazhong Agricultural University, Honshan District, Wuhan, Hubei Province, China (30°47′N, 114°34′E). The field experiment involved four different treatments: Control treatment (CS, soil without fertilizer), control organic fertilizer treatment (COF; 500 g/m^2; soil applied with COF), BSF organic fertilizer treatment (BOF; 500 g/m^2; soil applied with BOF) and BSF microbes agents organic fertilizer treatment (BMOF; 500 g/m^2; soil applied with BMOF). Each treatment consisted of three replicate plots (length: 2.6 m × width: 1.1 m). Cherry radish seeds were purchased from Botou Yonghong Seed Co., LTD. Cherry radishes were harvested in the field after 90 days, and rhizosphere soil was collected for further analysis. Soil and radish samples were independently collected from each plot to ensure the independence and reliability of the replicates. The cherry radishes from the CS, COF, BOF, and BMOF treatments were designated as CSC, COFC, BOFC, and BMOFC, respectively.

2.4. Metagenomic sequencing and binning analysis

The soil and cherry radish DNA were extracted using the Soil DNA Kit (Omega Bio-Tek, USA) according to previous research (Fan et al., 2023; Zhang et al., 2019, 2022b). The concentrations of the DNA were measured using a Nanodrop 2000 spectrophotometer (Thermo Scientific, USA) and the samples were subsequently stored at −80 °C for further analysis. These DNA samples were then subjected to metagenomic sequencing utilizing the Illumina NovaSeq 6000 platform.

The raw sequencing data underwent quality control and filtering using Trimmomatic, and subsequent removal of host sequences was accomplished using Bowtie 2. All radish and black soldier fly DNA were identified with reference to the radish (NC_018551.1) and black soldier fly (PRJEB37575) genomes and removed. The MEGAHIT software was utilized to assemble the cleaned reads following de-host gene processing in order to generate contigs (Li et al., 2015). Gene sequences within the contigs were predicted using Prodigal software (Hyatt et al., 2010). Additionally, the Cd-hit software was applied to obtain de-redundant genes (Li and Godzik, 2006). The Salmon software was utilized to quantify the de-redundant gene sequences (Patro et al., 2017).

The DIAMOND software was utilized to perform a blast and annotation of the de-redundant genes against the comprehensive antibiotic resistance database (CARD), thereby acquiring annotation information regarding potential antibiotic resistance genes present in each sample (Buchfink et al., 2015). Virulence factors (VFs) were identified through reference to the Virulence Factor Database (VFDB) (Liu et al., 2022). The MGEs database was used to further classify MGEs (Pärnänen et al., 2018). We conducted a further investigation into the co-occurrence of ARGs, VFs, and MGEs to ascertain whether these genetic components were co-located on the same contig. Potential pathogenic antibiotic-resistant bacteria (PARB) were defined as MAG that contains ARGs, MGEs, and VFs simultaneously.

Metagenomic binning was performed using Metabat2 (Gabrielli et al., 2023; Shaw and Yu, 2024; Tan et al., 2024), focusing on contigs exceeding 1500 bp in length for the binning process. Subsequently, the resultant bins were refined with RefineM to eliminate contigs exhibiting high levels of contamination (Parks et al., 2017). The completeness and contamination levels of the bins were assessed by using CheckM (Parks et al., 2015). Following this, dRep software was subsequently utilized for bin de-redundancy, selecting bins that exhibited over 75 % completion and less than 25 % contamination for further analysis (Olm et al., 2017).

2.5. Statistical analysis and visualization

Statistical significance was assessed using SPSS 25.0 through one-way ANOVA. The generation of heatmaps was conducted utilizing the “pheatmap” package in R Studio (version 4.3.1). The Mantel test was performed by the “vegan” package within R Studio. Variation partitioning analysis (VPA) was conducted with CANOCO 5.0. Pearson correlation analysis was carried out employing the “psych” package in R. Network analysis was performed in R Studio, with visualization aided by Gephi (version 0.10.1).

3. Results and discussion

3.1. Diversity and dynamics of ARGs in rhizosphere soil

A total of 191 ARGs subtypes were identified across all rhizosphere soil samples, with aminoglycoside, tetracycline, macrolide, and multidrug resistance genes emerging as the most prevalent. Notably, unfertilized soil exhibited the lowest diversity, comprising only 114 ARG subtypes. Following the application of COF, the number of ARG subtypes present in the rhizosphere soil increased to 153 (Fig. 1A). Similarly, the application of chicken manure organic fertilizer resulted in the detection of 188 types of ARGs in the soil (Zhu et al., 2022). The richness of ARGs

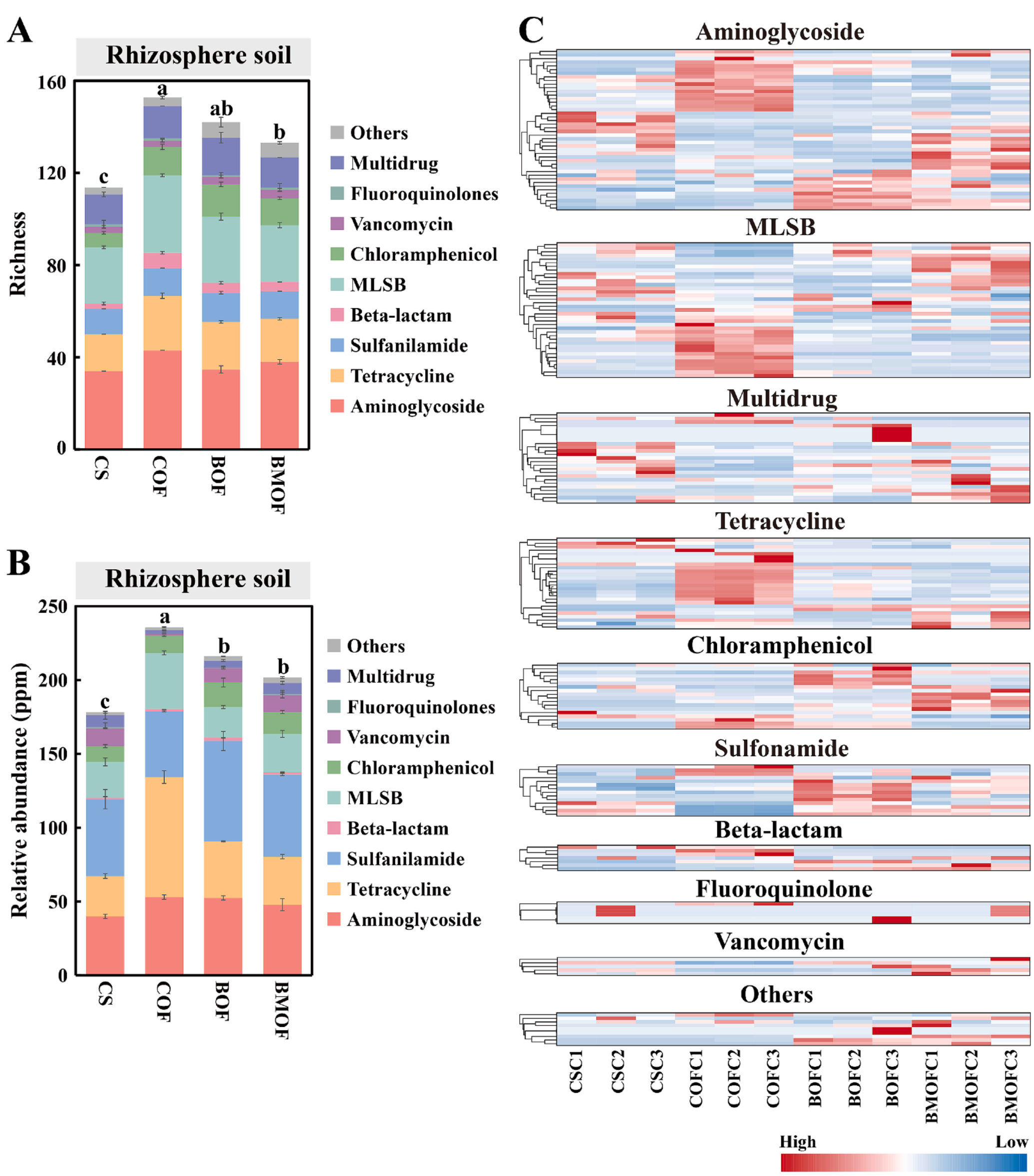

Fig. 1. Diversity and dynamics of ARGs in fertilized rhizosphere soil. **(A)** The richness of ARGs in fertilized rhizosphere soil. **(B)** Dynamics of ARGs in fertilized rhizosphere soil. **(C)** Heatmaps shows the relative abundance of different classes ARGs in rhizosphere soil.

in the rhizosphere soil of watermelon exhibited a significant increase following the application of organic fertilizer, with a total of 137 ARGs identified (Du et al., 2022). In contrast, the application of BOF resulted in a reduction of 11 subtypes of ARGs, while the application of BMOF led to a significant decrease of 20 subtypes of ARGs compared to the COF treatment.

In terms of the abundance of ARGs, unfertilized soil exhibited the lowest levels at 177.1 ppm, whereas soil treated with COF demonstrated the highest relative abundance at 235.7 ppm (Fig. 1**B**). A *meta*-analysis encompassing 94 prior studies demonstrated a significant 5.9-fold increase in the overall abundance of ARGs in soil following the application of manure in agricultural settings (Zhang et al., 2022a). The abundance of tetracycline and sulfonamide resistance genes in soil subjected to long-term organic fertilizer application was found to be 3×10^5 and 1×10^5 times greater than that observed in unfertilized soil, respectively (Zhao et al., 2019). Moreover, the application of BOF and BMOF resulted in significant reductions in the relative abundance of ARGs by 8.4 % and 14.4 %, respectively, in comparison to COF, with corresponding levels of 215.9 ppm and 201.7 ppm. Furthermore, the application of BSF fertilizer demonstrated particularly effective results in reducing the prevalence of tetracycline and MLSB resistance genes. In comparison to COF, the relative abundance of tetracycline resistance genes in the rhizosphere soil decreased by 52.9 % and 59.9 % subsequent to the application of BOF and BMOF, respectively. Meanwhile, the abundance of MLSB resistance genes exhibited reduction of 45.3 % and 32.21 %, respectively (Fig. 1**B,** Fig. 1**C**). Previous studies have demonstrated that BSF bioconversion can degrade 97 % of tetracycline and 95 % of tetracycline resistance genes in chicken manure (Cai et al., 2018a,b). The low levels of antibiotic residues and the antibiotic resistance in BSF fertilizer may contribute to the reduction of ARG richness and abundance in soil.

3.2. Diversity and dynamics of ARGs in cherry radish

A total of 72 subtypes of ARGs were identified in cherry radish, with a predominant presence of aminoglycoside resistance genes **(**Fig. 2**A)**. In contrast, Wang et al. (2024) identified 32 ARGs in grapefruit (Wang et al., 2024a). Root vegetables, due to their direct contact with soil, exhibit a higher susceptibility to ARG enrichment compared to leafy vegetables and fruits. Additionally, a high diversity of multi-drug resistance genes (*smeE*, *smeD*, *MexB*), sulfonamide resistance genes (*sul1*, *sul2* and *sul4*), and tetracycline resistance genes (*tet30*, *tet42*) was observed. Moreover, the vancomycin resistance gene *vanRO* was identified in both COF and BOF radish samples. Vancomycin is utilized as a last-resort therapeutic agent for the management of severe gram-positive bacterial infections in clinical contexts (Mühlberg et al., 2020), thereby highlighting the clinical importance of addressing antibiotic resistance in plant systems. Additionally, a total of 38 subtypes of ARGs were identified in COF radish. The application of BOF and BMOF resulted in a reduction of ARGs subtypes in radish by 13 and 10, respectively, in comparison with COF **(**Fig. 2**A, B)**. Further analysis indicated that 28 subtypes of ARGs could migrate from soil to radish through the application of COF. In contrast, the applications of BOF and BMOF significantly reduced the migration risk of 9 and 6 ARGs from soil to radish, respectively **(**Fig. 2**C, D; Table. S1)**.

The relative abundance of ARGs in radish samples from CS, COF, BOF, and BMOF was measured at 4.4 ± 0.6 ppm, 7.6 ± 1.2 ppm, 5.5 ± 1.0 ppm, and 5.1 ± 0.5 ppm, respectively. Notably, the BOF and BMOF treatments resulted in a reduction of ARGs in radish by 27.6 % and 32.4 %, respectively, in comparison to the COF treatment **(**Fig. 2**E)**. The aminoglycoside resistance gene *parY* exhibited the highest abundance across all samples, indicating its predominance in radish populations. In contrast, Guo et al. (2021) identified 150 ARGs across four types of leafy vegetables (coriander, lettuce, bok choy, and lettuce leaves). Their findings indicated that the leaves of lettuce and bok choy exhibited a higher concentration of ARGs compared to the leaves of coriander and lettuce leaves. Notably the genes *aadE*, *tet34*, and *vanSB* were prevalent across all four vegetable types. Kläui et al. (2024) conducted an analysis of 150 local and imported samples of fresh tomatoes, carrots, lettuce, strawberries, and cilantro, revealing that the absolute abundance of ARGs ranged from 10^4-10^6 copies per gram. Importantly, the genes *sul1* and *blaTEM* were detected in more than 60 % of the examined samples. The presence of ARGs in crop samples is influenced by a variety of factors, including plant species, soil fertilization practices, soil physicochemical properties, sources of irrigation water, and environmental pollutants.

In comparison to COF, the application of BOF resulted in a reduction exceeding 50 % in the relative abundance of 33 subtypes of ARGs, with 20 of these ARGs becoming undetectable. Specifically, the relative abundance of resistance genes associated with beta-lactam, tetracycline, vancomycin, and MLSB decreased by 100 %, 95.4 %, 85.2 %, and 84.6 %, respectively. In contrast, following the application of BMOF, the relative abundance of 32 ARGs in radish decreased by more than 50 %, with 16 ARGs becoming undetectable. Notably, the relative abundance of resistance genes for vancomycin, beta-lactam, MLSB, and chloramphenicol decreased by 100 %, 99.2 %, 92.9 %, and 92.7 %, respectively (Fig. 2**E, F**). These results suggest that applying BSF organic fertilizer may effectively mitigate the risk of ARGs migration from soil to plants.

3.3. MGEs, VFs and their co-occurrence with ARGs in cherry radish

ARGs can disseminate among bacterial populations through horizontal gene transfer (HGT), which is mediated by mobile genetic elements (MGEs) (Wang et al., 2024b). The relative abundance of MGEs in the BOF and BMOF radish exhibited a significant reduction of by 32.8 % and 65.8 %, respectively, compared to the COF **(**Fig. 3**A)**. Transposases represent the predominant MGEs in radish, and their relative abundance in the BOF and BMOF treatments exhibited a decrease of 59.7 % and 86.8 %, respectively, when compared to the COF, respectively **(**Fig. 3**A)**. Previous studies have identified transposases as the primary MGEs in tomatoes, facilitating the HGT of ARGs (Sun et al., 2021). Additionally, another study also found that 97.2 % of MGEs in chicken meat are transposases, underscoring their role as high-risk carriers for the transfer of ARGs across diverse environmental samples (Vieira et al., 2023). Further analysis utilizing Pearson's correlation-based linear regression revealed a statistically significant positive correlation between MGEs and ARGs ($r = 0.77$, $p = 0.0036$, Fig. 3**B**). Moreover, LEfSe identified differentially abundant MGE biomarkers, specifically *tnpA_IS683*, *IS91*, and *tnpA*, which were detected at COFC, BOFC, and BMOFC, respectively (Fig. 3**C; Fig. S1**).

The relative abundance of VFs in the BOF and BMOF radish was observed to decrease by 29.1 % and 59.4 %, respectively, in comparison to the COF **(**Fig. 3**D)**. Adherence and mobility were the predominant VFs, accounting for 41.3 %% to 51.4 % of the observed metrics. Notably, the relative abundance of biofilm in the BMOF was significantly diminished, exhibiting a reduction of 89.2 % compared to that in the COF. Biofilms formed by antibiotic-resistant bacteria serve to protect them from external disturbances, thereby exacerbating the dissemination of ARGs (Yu et al., 2024). Both previous research (Kim et al., 2022) and the current study (Pearson correlation, $r = 0.66$, $p = 0.02$, Fig. 3**E**) indicate a significant positive correlation between ARGs and VFs. Furthermore, biomarker VFs including VFG016490 (EF-Tu), VFG043550 (GroEL), VFG042942 (Type IV pili), and VFG046459 (EF-Tu) were identified at CSC, COFC, BOFC and BMOFC, respectively (Fig. 3**F**). Collectively, these results suggest that BSF fertilizer may mitigate the risk of horizontal transfer of ARGs and reduce the prevalence of ARG-carrying pathogens in radish.

3.4. Bacterial and fungal community structures in cherry radish

At the phylum level, fungi exhibited a higher prevalence across all radish samples, with their relative abundance ranging from 43.3 % to 57.8 % **(Fig. S2; Table S2)**. In contrast, bacteria displayed a relative

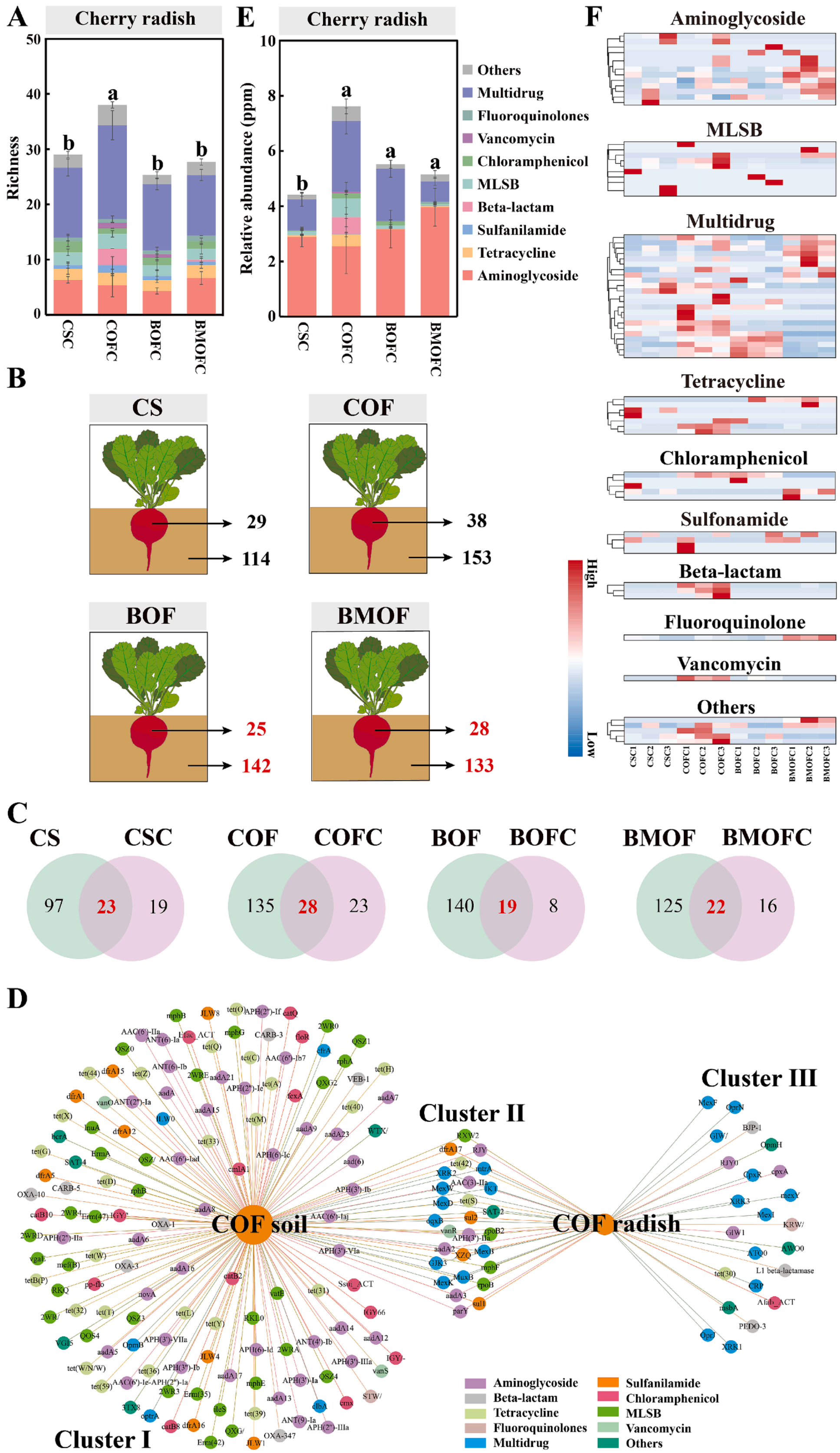

Fig. 2. Diversity and dynamics of ARGs in cherry radish. **(A)** The richness of ARGs in cherry radishes. **(B)** Number of ARGs detected in the rhizosphere soil and cherry radishes in each treatment. **(C)** Number of ARGs shared in rhizosphere soil and cherry radishes in different fertilization treatments. The red letters denote the quantity of ARGs that are shared between the soil and the radish. This figure represents the number of ARGs that have the potential to be transferred to the radish via fertilization. **(D)** Network analysis illustrates shared and unique ARGs in rhizosphere soil and cherry radish in COF treatment. Cluster II highlights ARGs that risk transferring from the soil to the cherry radish. **(E)** Dynamics of ARGs in cherry radish. **(F)** Heatmaps show the relative abundance of ARGs in cherry radish.

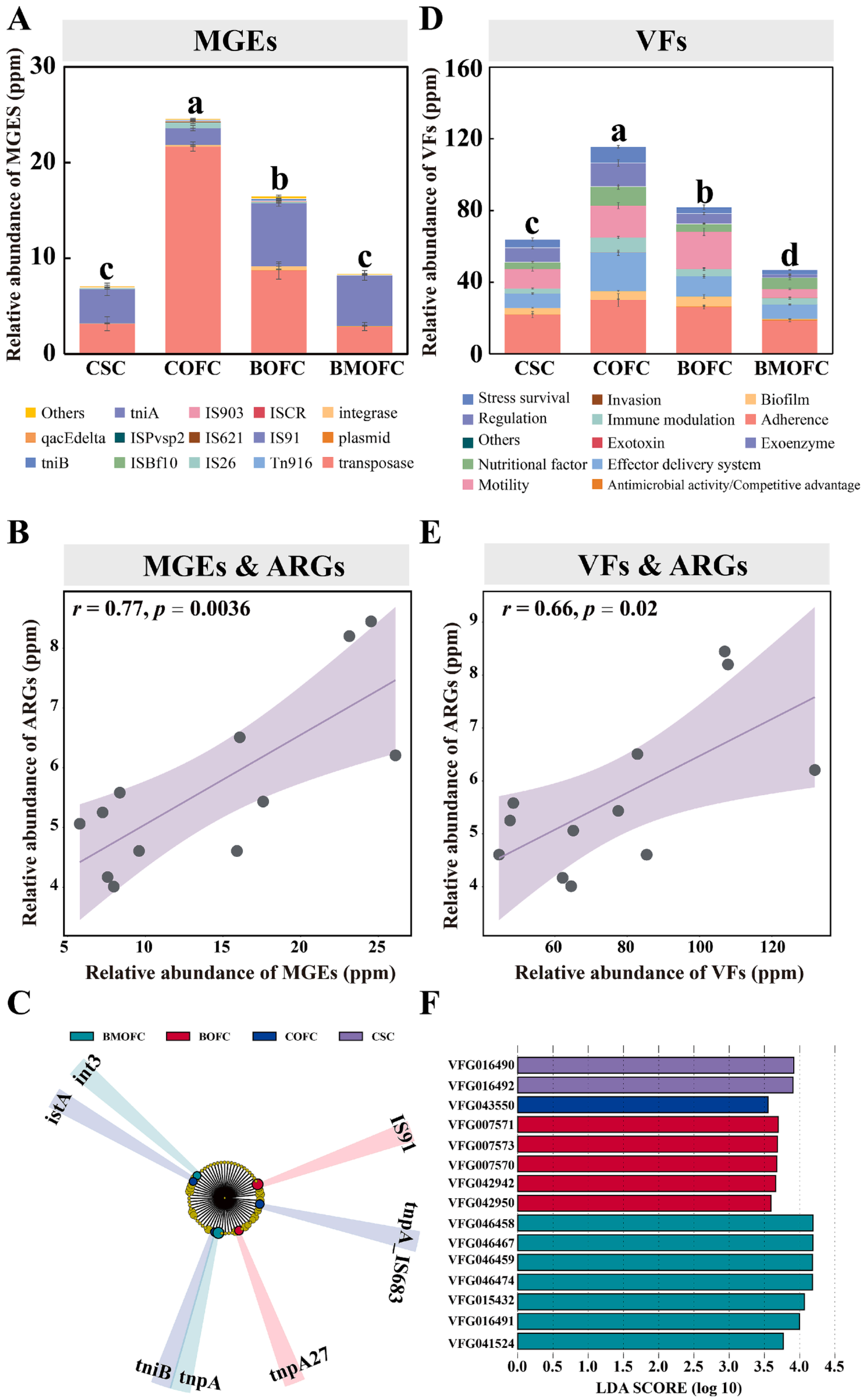

Fig. 3. MGEs, VFs and their co-occurrence with ARGs. **(A)** Relative abundance of MGEs in different radish samples. **(B)** Linear regression based on Pearson's correlation indicates a significant positive correlation between MGEs and ARGs. **(C)** LEfSe (Linear Discriminant Analysis effect size) was employed to identify differential MGEs (biomarkers). Biomarkers were detected via LDA effect size with criteria of LDA > 3.5 and p < 0.05. **(D)** Relative abundance of VFs in different radish samples. **(E)** Linear regression based on Pearson's correlation indicates a significant positive correlation between VFs and ARGs. **(F)** LEfSe (Linear Discriminant Analysis effect size) was employed to identify differential VFs (biomarkers). Biomarkers were detected via LDA effect size with criteria of LDA > 3.5 and p < 0.05.

abundance ranging from 17.3 % to 39.7 % (Fig. 4A; Table S3). Among the various fungal phyla, Chytridiomycota emerged as the most dominant, comprising between 29.3 % and 39.4 % of the total fungal community, this was followed by Ascomycota, which accounted for 13.3 % to 17.7 %. Pseudomonadota represented the most abundant group, exhibiting a range of 10.2 % to 26.4 %, while Cyanobacteriota constituted 4.9 % to 5.6 %. At the genus level, *Synchytrium* was identified as the predominant fungal group, with a relative abundance ranging from

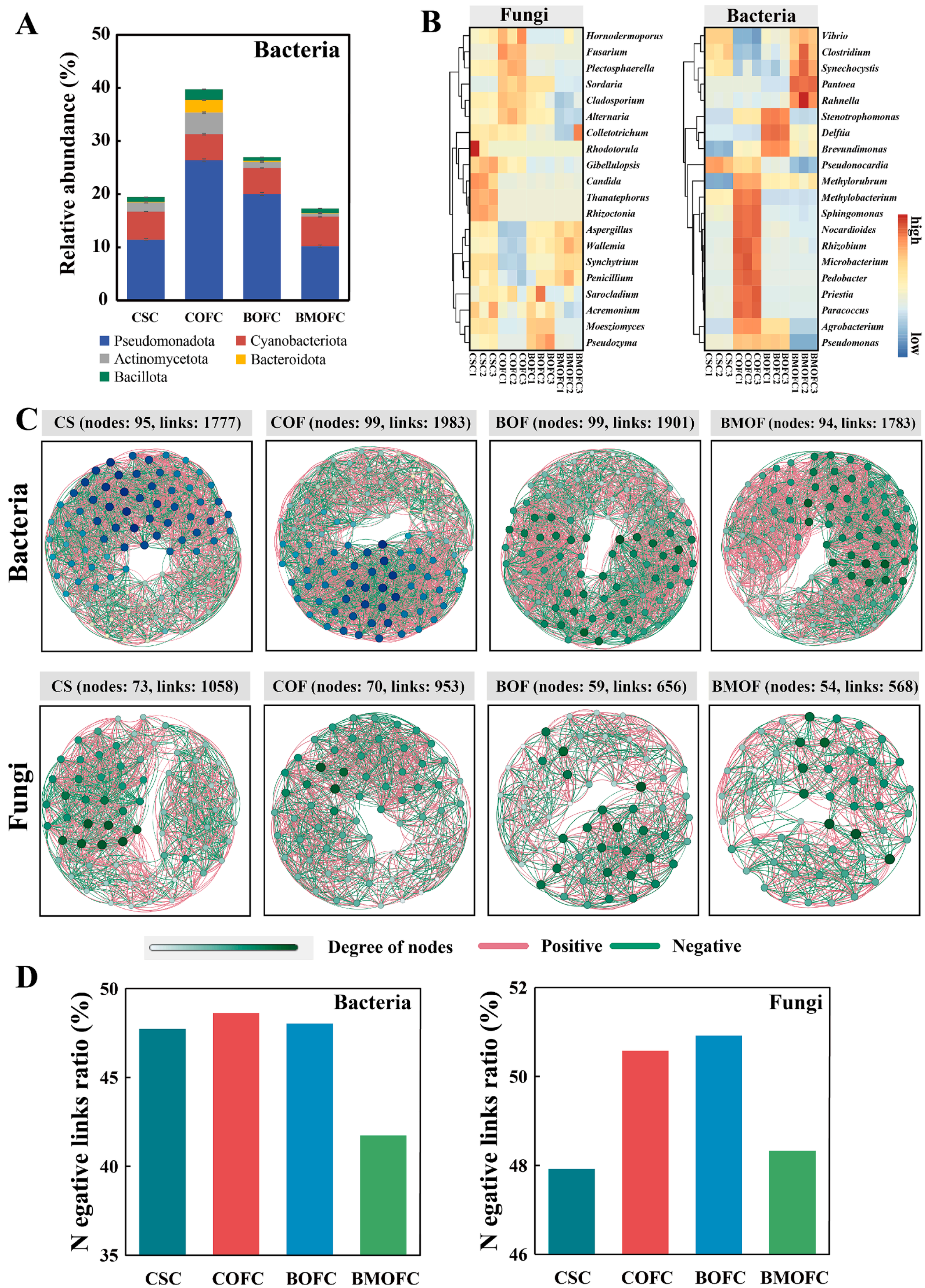

Fig. 4. Bacterial and fungal community structures in cherry radish. **(A)** Relative abundance of the TOP 5 bacterial phyla. **(B)** Heatmaps illustrate the relative abundance of the TOP20 bacterial and fungal at genus level. **(C)** Molecular ecological network of bacteria and fungi was constructed using network analysis across various treatments. Nodes within the network represent bacterial or fungal taxa at the genus level. The size of each node reflects its degree of connectivity, while edges denote the relationships between nodes. These connections were determined based on a Pearson correlation coefficient threshold of $r > 0.6$ and a significance level of $p < 0.05$. **(D)** Proportion of negative links in bacterial and fungal molecular ecological networks.

29.1 % to 39.8 % (Fig. 4B). *Aspergillus* was identified as the second most abundant genus, contributing between 10.9 % and 17.8 % to the total fungal population. Regrading bacterial composition, *Synechocystis* emerged as the most prevalent genus, exhibiting an abundance ranging from 4.8 % to 5.7 %, this was followed by *Stenotrophomonas*, which displayed an abundance of 0.3 % to 5.7 %. In contrast to the bacterial and fungal communities, the relative abundance of archaeal communities was notably lower, accounting for only 0.3 %-0.4 % of the total microbial population across all samples. Among the archaeal genera, *Halarchaeum* was the predominant genus, followed by *Haloterrigena*, *Haloferax*, and *Candidatus Nitrosocosmicus* (**Fig. S3**).

Microbial molecular ecological network analysis indicated that, despite the lower abundance of bacteria in radishes, their interaction strength was significantly greater, exhibiting between 1777 and 1983 links, compared to fungi, which displayed between 568 and 1058 links (Fig. 4C). In the BMOF group of radishes, the proportion of negative links within both bacterial and fungal networks was notably lower compared to other groups, recorded at 41.7 % for bacteria and 48.3 % for fungi (Fig. 4D). This finding indicates a predominance of synergistic interactions over competitive interactions among the microorganisms present in BMOF radishes.

3.5. Correlation analysis of MGEs, VFs and microbial communities with ARGs in cherry radish

Network analysis revealed the co-occurrence of ARGs, MGEs, VFs, and microbial communities in cherry radishes (Fig. 5A), suggesting a potential interplay among these elements. Furthermore, Mantel tests indicated significant correlations between MGEs and VFs with specific ARGs, including those conferring resistance to tetracycline, chloramphenicol, vancomycin, and multidrug (Fig. 5B). Previous studies have identified bacterial communities as significant contributors to the prevalence of ARGs (Wang et al., 2024b; Zhu et al., 2023). Our variation partitioning analysis (VPA) further elucidates this relationship, demonstrating that bacteria, VFs, and MGEs account for 13.0 %, 12.7 %, and 11.0 % of the variation in ARG patterns, respectively (Fig. 5C). This finding underscores the pivotal role of bacteria, as opposed to fungi, in shaping the profiles of ARGs.

Additionally, we evaluated the complexity of molecular ecological networks encompassing MGEs, VFs and microbial communities associated with ARGs in cherry radish. The network analysis of ARGs and MGEs revealed a significant reduction in the number of links was observed in the BOF and BMOF treatments, with decreases of 54.2 % and 32.8 %, respectively, compared to the COF treatment (Fig. 5D; **Fig. S4**). Similarly, the networks of ARGs and VFs demonstrated a reduction in the number of connections in both BOF and BMOF, with decreases of 32.3 % and 43.4 %, respectively, in comparison to COF (Fig. 5E; **Fig. S5**). Finally, the network analysis of ARGs and microbial communities demonstrated a reduction in the number of connections, with decreases of 32.8 % and 23.0 % observed in the BOF and BMOF treatments, respectively, compared to the COF (Fig. 5F; **Fig. S6**). These results indicate a reduction in network complexity in both BOF and BMOF treatments compared to the COF treatment, suggesting a potential modification in the ecological interactions among ARGs, MGEs, VFs, and microbial communities under varying fertilization treatments.

3.6. MAG analysis reveals the co-occurrence of ARGs, MGEs, and VFs in cherry radish

Metagenomic-assembled genomes (MAGs) derived from all radish samples resulted in the assembly of 117 MAGs. With the exception of Bin COFC2.15, which was annotated as Bacteroidota, and Bins COFC2.6 and COFC2.17, which were annotated as Actinomycetota, along with those that were not annotated to specific species, the majority of the remaining MAGs were predominantly classified under Pseudomonadota, with a substantial proportion identified as *Sphingomonas* (representing 47.4 % of the annotated MAGs, Fig. 6A).

A total of 91 MAGs were annotated with ARGs in radish samples. Notably, ARGs, MGEs, and VFs co-occurred in 15 MAGs (**Table S4**), indicating that these MAGs may represent potential PARB, which pose a risk for the dissemination of ARGs. Further analysis revealed that, compared to COF, the abundance of PARB in BOF and BMOF radishes decreased by 45.6 % and 94.0 %, respectively (Fig. 6B).

Further analysis revealed the coexistence of ARGs and MGEs in genetic context (Fig. 6C). For instance, the insertion sequences *ISAzo26* and *ISMo24* were identified as coexisting with multidrug resistance genes *mdtC*, *mdtA*, *norM*, and fluoroquinolone resistance genes *emrA* and *emrB* in Bin COFC2.8 Contig_118153 (Fig. 6D). Furthermore, transposase *TnXax1* was observed to coexist with multidrug resistance genes *mdtA* and *mdtC* on Bin COFC2.18 Contig_52999. Insertion sequences *ISSme1* and *ISAtu5* were identified as coexisting with tetracycline resistance genes *tetA* and *tetR*, as well as the multidrug resistance gene *mdtL*.

Evidence of genetic context revealing the coexistence of ARGs, MGEs, and VFs was also identified. In Bin COFC2.18 Contig_179027, the insertion sequence *ISHne3* was found to coexist with multidrug resistance genes *mdtC* and biofilm-associated virulence factors *bdcA* (Cyclic-di-GMP-binding biofilm dispersal mediator protein) and *bigR* (Biofilm growth-associated repressor). Notably, multidrug resistance genes *mdtC* and *bmrA*, along with virulence factors associated with flagella and biofilm formation, have been identified in the opportunistic pathogen *Stenotrophomonas maltophilia*.

3.7. Mechanisms underlying the reduced ARGs transfer risks from fertilized soil to radish by BOF application

Soil antibiotic resistance poses a significant environmental threat to human health, primarily through the transfer of ARGs to crops and their subsequent entry into the food chain. Although prior research has extensively explored methods for removing ARGs from soil and mitigating their dissemination, the impacts of ARGs migration to crops remain poorly understood. In this study, we focus on the application of BOF. While previous studies have demonstrated that BOF reduces the abundance of ARGs in fertilized soil compared to control conditions (Zhao et al., 2024), the effects of BOF on ARGs in vegetables have not been thoroughly investigated. Through metagenomic sequencing and field experiments, our study demonstrates that BOF application can effectively reduce the risk of ARG migration to vegetable. This finding provides novel strategies and robust scientific evidence for mitigating the transfer risks of ARGs into the human food chain.

Moreover, our study introduces functional fungi and bacteria that can antagonize antibiotic-resistant bacteria into BOF to enhance their effectiveness in reducing ARGs in soil and plants. This innovative approach not only improves the efficiency of ARG mitigation but also provides scientific support for sustainable agricultural health management.

Furthermore, the application of BOF significantly mitigates the transfer risks of ARGs from fertilized soil to radish through a multifaceted mechanism.

Firstly, our previous studies have demonstrated that the abundance of ARGs in BOF is significantly lower than that in COF (Cai et al., 2018a; Zhao et al., 2024). Therefore, BOF application could reduces the exogenous input of ARGs, thus mitigating the risk of ARGs entering the soil and migrating to radish. Moreover, BOF is rich in beneficial microorganisms, many of which originate from the BSF gut, such as *Bacillus* species (Zhao et al., 2023a, 2024). These beneficial microorganisms can effectively antagonize antibiotic – resistant bacteria (ARB) in the soil. Since ARB serve as hosts of ARGs, a reduction in ARB abundance directly leads to a decrease in ARGs abundance (Ding et al., 2023). This antagonistic effect further limits the potential for ARGs to be transferred from the soil to radish plants.

Meanwhile, BSF larvae have been shown to effectively degrade

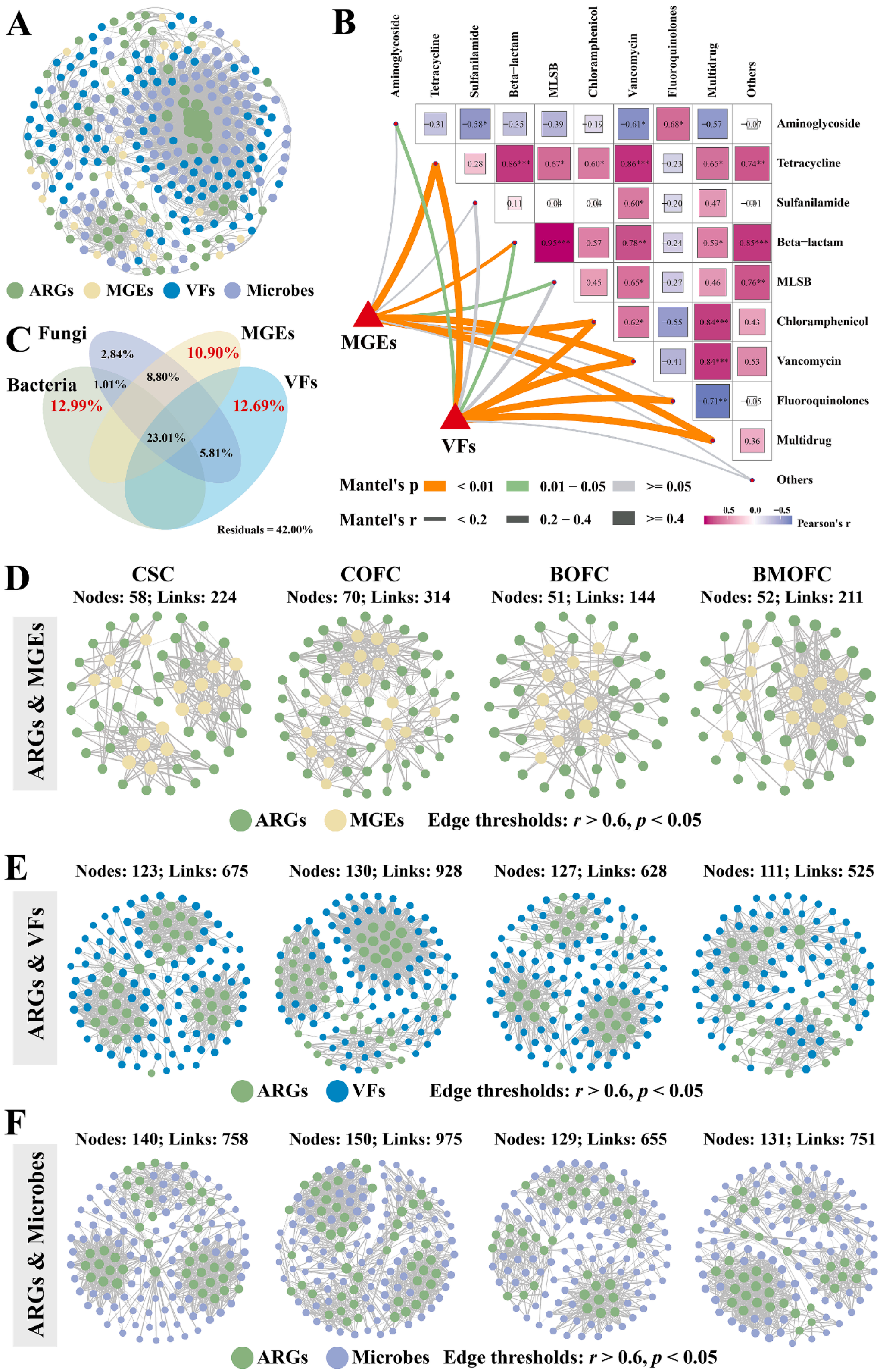

Fig. 5. Correlation analysis of MGEs, VFs and microbial communities with ARGs in cherry radish. **(A)** Network analysis reveals the co-occurrence of ARGs, MGEs, VFs and microbial communities. **(B)** Mantel tests demonstrate the correlation between MGEs and VFs with ARGs. The width of the edges corresponds to Mantel's r statistic for the respective distance correlations, while the color of the edges signifies statistical significance based on 9,999 permutations. Pairwise comparisons among different classes of ARGs are illustrated with a color gradient representing Spearman's correlation coefficients. **(C)** Variation partitioning analysis (VPA) elucidates the contributions of MGEs, VFs and microbial communities (including bacteria and fungi) to the distribution of ARGs. **(D)** Network analysis reveals the co-occurrence of MGEs and ARGs in different treatments. **(E)** Network analysis reveals the co-occurrence of VFs and ARGs in different treatments. **(F)** Network analysis reveals the co-occurrence of microbial communities and ARGs across various treatments. The size of each node corresponds to its connectivity, with edges representing the relationships between nodes. These connections are identified based on a Pearson correlation coefficient threshold of $r > 0.6$ and a significance level of $p < 0.05$.

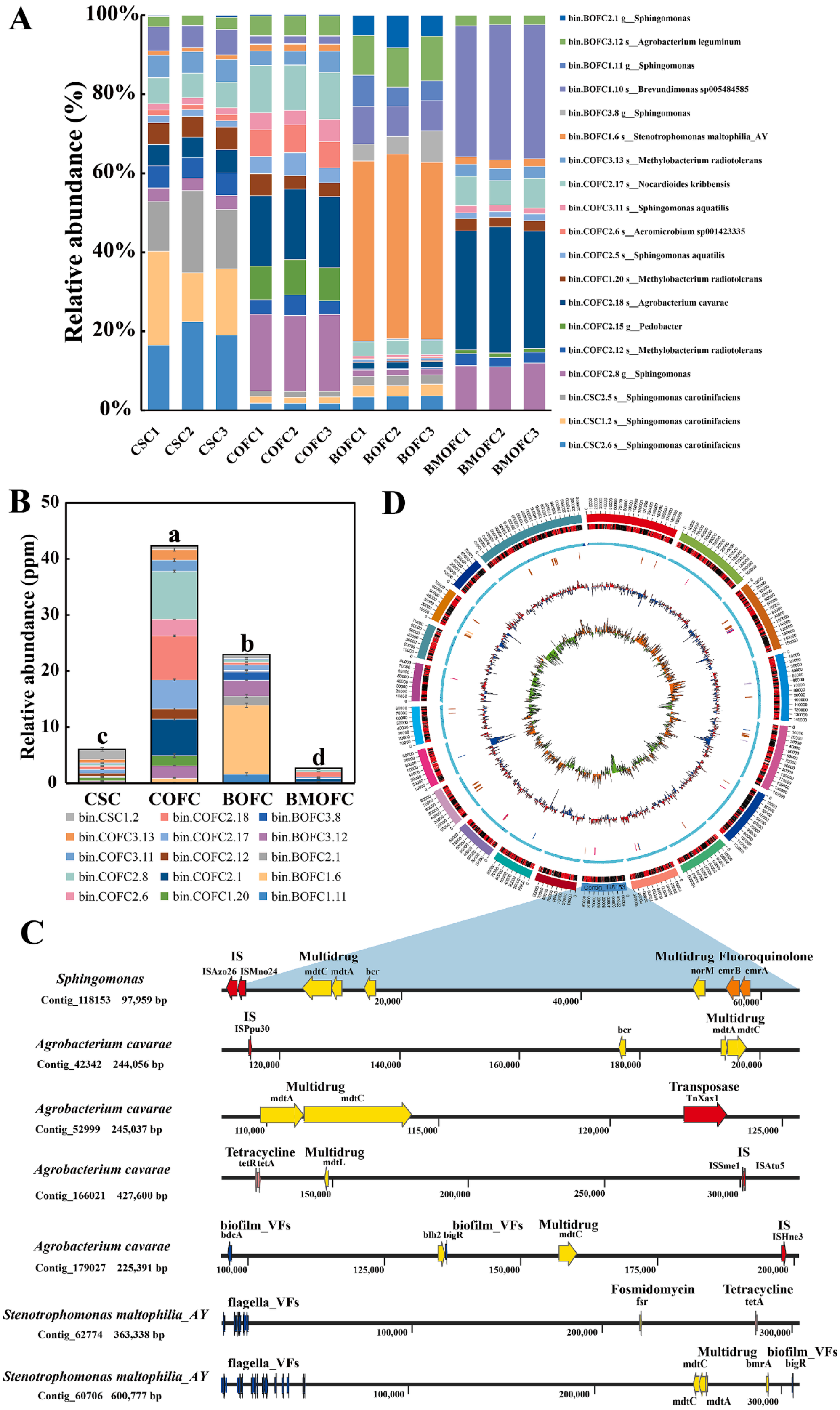

Fig. 6. Metagenomic-assembled genomes (MAGs) analysis reveals the co-occurrence of ARGs, MGEs, and VFs in cherry radish. **(A)** Relative abundance of MAGs across different treatments. **(B)** Relative abundance of MAG carrying ARGs, MGEs and VFs simultaneously. **(C)** Genetic structures of ARGs, MGEs, and VFs in the same contig. **(D)** Cricos plot of Bin COFC2.8.

antibiotics in manure (Li et al., 2025). This degradation reduces the concentration of antibiotics in the soil, consequently decreasing the associated antibiotic selection pressure. As a result, the risk of the development and dissemination of antibiotic resistance is significantly mitigated. Furthermore, the application of BSF larvae application enhances soil fertility and increases enzyme activity (Lopes et al., 2022). These alterations in the soil micro-environment may influence the succession of soil microbial community structures. A more balanced and diverse microbial community could outcompete ARB, thereby reducing the overall abundance of ARGs in the soil.

Furthermore, the application of BOF may reduce the abundance of MGEs in soil. MGEs play a critical role in the horizontal transfer of ARGs among bacteria (Meng et al., 2022). By decreasing the prevalence of MGEs, BOF application effectively reduces the risk of ARGs spreading and transferring to radish plants through soil microbial networks. Lastly, BOF application enhances plant root development and stress tolerance (Raman et al., 2024). Stronger and healthier root systems may reduce the uptake of ARGs by radish plants, either by limiting the availability of ARGs in the rhizosphere or by improving the plant's ability to resist microbial colonization. This indirect effect further contributes to the reduced transfer of ARGs from soil to radish.

These mechanisms function synergistically to effectively mitigate the transfer of ARGs from soil to radish, thereby presenting a promising approach for reducing antibiotic resistance risks within agricultural ecosystems.

3.8. Limitations and future research directions

In this study, field experiments were conducted over a duration of 90 days. Future research should incorporate longer-term field trials to examine the dynamics of ARGs and their influencing factors over extended periods. Additionally, the impact of environmental factors warrants further consideration. For example, fluctuations in climate and precipitation across different regions may affect the efficacy of BOFs on ARGs. Subsequent studies should investigate these effects. Furthermore, this study primarily focuses on cherry radish; however, distinct types of vegetables may demonstrate varying capacities to absorb, accumulate, and transfer ARGs from the soil. There is a need for more research on a diverse range of plant species. Moreover, while this study explores the transfer of ARGs from fertilized soil to radish, it provides limited direct evidence regarding the potential risks to human health. Future investigations could include experiments involving mice or humans to validate the risks associated with the full food chain transfer of ARGs. Additionally, conducting experimental validations of the potential for horizontal gene transfer or pathogenicity would be advantageous. Lastly, soils frequently contain multiple pollutants, such as heavy metals, pesticides, and microplastics, which may interact with ARGs. Future research should examine the comprehensive impact of BOFs on ARGs in the presence of other pollutants.

4. Conclusions

Overall, our study demonstrates that BSF larvae can efficiently bioconvert animal manure into organic fertilizer, and the application of BSF organic fertilizer significantly reduced the transfer risks of ARGs from fertilized soil to cherry radish. In comparison to the COF treatment, BOF significantly mitigated the abundance of ARGs, MGEs, and VFs. The study also revealed significant correlations among MGEs, VFs, and ARGs, with the BOF treatment notably decreasing their relative abundance in radish. Metagenomic analysis indicated the co-occurrence of ARGs, MGEs, and VFs, suggesting the presence of pathogenic antibiotic-resistant bacteria in radish. Notably, the abundance of PARB was significantly reduced in the BOF treatment. These findings underscore the potential of BOF in mitigating transfer risks associated with ARGs and reducing pathogen presence in radish, thereby emphasizing the significance of sustainable agricultural practices.

CRediT authorship contribution statement

Zhengzheng Zhao: Writing – original draft, Visualization, Methodology, Formal analysis, Data curation. **Bingqi Gao:** Formal analysis, Data curation. **Ahmed R. Henawy:** Writing – review & editing. **Kashif ur Rehman:** Writing – review & editing. **Zhuqing Ren:** Writing – review & editing. **Núria Jiménez:** Writing – review & editing. **Longyu Zheng:** Writing – review & editing. **Feng Huang:** Writing – review & editing. **Ziniu Yu:** Writing – review & editing. **Chan Yu:** Funding acquisition, Formal analysis. **Jibin Zhang:** Writing – review & editing, Funding acquisition. **Minmin Cai:** Writing – review & editing, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envint.2025.109510.

Data availability

All sequencing data for this study are available from the NCBI database (Accession number: PRJNA1240223).

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